

A-Share Multi-Factor Quantitative Research

Out-of-sample factor validation, portfolio backtesting and risk analysis

TRAINED
FACTOR
RETURN

7.53%

CSI 300
RETURN

16.40%

SHARPE RATIO

0.20

MAXIMUM
DRAWDOWN

-21.34%

Executive Summary

The trained multi-factor model returned **7.53%** during the true out-of-sample period. The original model returned **-1.08%**, while the CSI 300 returned **16.40%**.

Training improved the original strategy by **8.61 percentage points**. However, the trained model produced an excess return of **-8.87 percentage points** relative to the benchmark.

Training improved performance relative to the original model. The trained factor model underperformed the CSI 300.

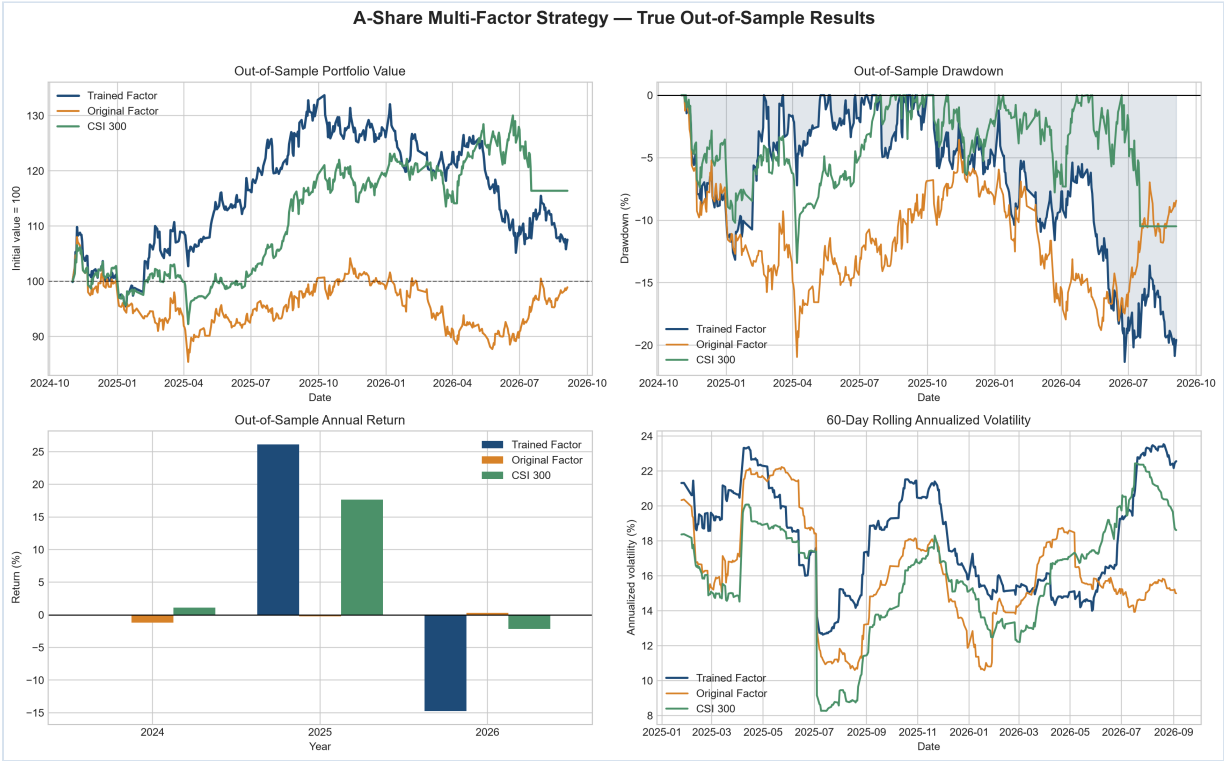
Research Methodology

- Twelve selected A-share companies and the CSI 300 benchmark.
- Six price- and volume-based factors.
- Cross-sectional winsorisation and Z-score standardisation.
- Monthly selection of the top five stocks.
- Equal weighting with a 20% stock limit.

- Transaction commission, stamp duty and slippage included.
- Chronological 70% training and 30% testing split.
- Factor weights frozen before the out-of-sample backtest.

True Out-of-Sample Performance

Method	Final Value	Total Return (%)	Annual Return (%)	Annual Volatility (%)	Sharpe Ratio	Maximum Drawdown (%)	Cost (%)
Trained Factor	107530.82	7.53	4.15	18.87	0.20	-21.34	1.46
Original Factor	98920.29	-1.08	-0.61	16.60	-0.07	-20.95	1.28
CSI 300 Benchmark	116398.19	16.40	8.88	16.33	0.48	-13.42	0.00



Factor Validation

Period	Months	Mean IC	IC Standard Deviation	IC IR	Positive IC Rate (%)	t-Statistic
Testing	21	0.03	0.39	0.08	61.90	0.37
Training	48	0.09	0.35	0.26	54.17	1.80

Trained Factor Weights

Factor	Training Mean IC	Testing Mean IC	Direction	Weight (%)	Stable Direction
20-Day Momentum	0.03	-0.03	1	15.63	0
60-Day Momentum	-0.04	-0.18	-1	17.45	1
5-Day Reversal	-0.04	0.04	-1	20.32	0
Low Volatility	-0.07	0.11	-1	31.53	0
MA Trend	-0.03	-0.16	-1	12.96	1
Volume Ratio	0.00	-0.00	1	2.10	0

Out-of-Sample Risk Analysis

Method	95% VaR (%)	95% CVaR (%)	Worst Day (%)	Maximum Drawdown (%)	Maximum Loss Streak	Monthly Win Rate (%)
Trained Factor	1.75	2.47	-5.17	-21.34	10	52.17
Original Factor	1.41	2.26	-6.32	-20.95	10	43.48
CSI 300 Benchmark	1.71	2.55	-7.05	-13.42	5	60.87

Portfolio Turnover

Method	Rebalances	Total Turnover (%)	Average Monthly Turnover (%)	Annualised Turnover (%)
Original Factor	21	920.0	43.81	525.71
Trained Factor	21	1040.0	49.52	594.29
CSI 300 Benchmark	0	0.0	0.00	0.00

Latest Factor Ranking

Latest available ranking date: 2026-09-04

Rank	Symbol	Company	Composite Score
1	000001.SZ	Ping An Bank	0.60
2	600036.SS	China Merchants Bank	0.54
3	000333.SZ	Midea Group	0.39
4	000651.SZ	Gree Electric	0.33
5	600900.SS	Yangtze Power	0.26

Research Conclusion

The trained factor specification improved upon the original model but did not outperform the CSI 300 during the out-of-sample period.

Its lower Sharpe ratio, larger drawdown and high annualised turnover indicate that the current factor signal is not sufficiently strong for live deployment.

The strategy should be treated as a quantitative research prototype. Future work should expand the stock universe, include fundamental factors, neutralise sector exposure and apply walk-forward validation.

Limitations

- Small stock universe and possible survivorship bias.
- Simplified transaction cost assumptions.
- No detailed suspension or price-limit model.
- No sector-neutral portfolio construction.
- Limited number of out-of-sample months.
- Results do not guarantee future returns.

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